Claims

Claim 1

A system for automatic multi-level revenue allocation of cognitive assets based on smart contracts, deployed on a **decentralized distributed right confirmation ledger**, characterized in that the system comprises:

1. A **dependency graph tree construction unit**: configured to, upon triggering of a sub-protocol node registration transaction, read parent protocol hash fingerprints declared in a data structure of the sub-protocol node, recursively construct a tree-shaped derived topology graph along version dependency records, and set a hard **maximum dependency level limit threshold**; when a calculated recursive path depth exceeds the limit threshold, an exception is automatically thrown and decentralized consensus chaining of the registration transaction is blocked;

2. A **state variable incremental calculation unit based on cascaded topology paths**: configured to, upon receiving a data packet of a commercial settlement event, intercept an asset liquidation request containing a transaction amount **R**, split the amount **R** into a first proportional value retained by a current node, a second proportional value allocated to a parent protocol layer, and a third proportional value allocated to a grandparent protocol layer that satisfy a preset asymmetric descending gradient relationship, and write each proportional value into corresponding **pending withdrawal balance state mapping variables** of contracts in a ledger state tree in parallel in the form of non-immediate transfer increments;

3. A **dynamic weight allocation module** deployed inside the state variable incremental calculation unit: configured to, when the sub-protocol node inherits multiple parents, read self-consistency similarity between feature vector fingerprints of each parent protocol, calculate allocation weights proportionally, and accordingly dynamically adjust pending withdrawal balance increments of corresponding parent creator accounts in the state variables;

4. A **lifecycle state transition contract module**: configured to store a creation timestamp of each protocol node in the ledger state tree; when the state variable incremental calculation unit allocates pending split amounts to a target node, if a difference between a current consensus timestamp and the creation timestamp of the node exceeds a set protection period time threshold, or the node is determined as a **root-level meta-protocol with zero private revenue**, a contract state machine transition is triggered to forcibly redirect and accumulate the split amount corresponding to the node into a pending withdrawal balance variable of an **ecological**

directional backfeed pool contract account;

5. A **secure pull-type withdrawal unit**: configured to respond to a transfer transaction initiated by a payee, verify pending withdrawal balances under the initiator's name in the state variables, clear the pending withdrawal balance state variable, and then trigger an underlying on-chain transfer instruction to execute physical transfer to the payee, so as to physically isolate and eliminate reentrancy calls.

Claim 2

The system for automatic multi-level revenue allocation of cognitive assets according to **Claim 1**, characterized in that:

The hard **maximum dependency level limit threshold** set in the dependency graph tree construction unit is fixed to **18 levels**; if a recursive depth between the sub-protocol node and the **root-level meta-protocol with zero private revenue** is greater than 18 levels, the consensus forcibly rejects chaining registration.

Claim 3

The system for automatic multi-level revenue allocation of cognitive assets according to **Claim 1**, characterized in that:

In the state variable incremental calculation unit based on cascaded topology paths, split proportion gradients corresponding to the first proportional value, the second proportional value and the third proportional value are hard-coded and locked as follows: **75% retained by the creator of the current node, 10% allocated to the parent protocol layer, and 15% allocated to the grandparent protocol layer**. In liquidation along the same cascaded path, the third proportional value allocated to the traced grandparent protocol layer is permanently higher than the second proportional value allocated to the parent protocol layer, so as to drive the proliferation of underlying public infrastructure via reverse benefit gravity.

Claim 4

A method for automatic multi-level revenue allocation of cognitive assets based on smart contracts, executed by a computer processor, characterized by comprising the following steps:

1. Topology graph registration and verification: upon triggering of a sub-protocol node registration transaction, read parent protocol hash fingerprints declared in the data structure of the sub-protocol node, recursively construct a tree-shaped derived topology graph along version dependency records, and set a hard maximum dependency level limit threshold; when a calculated recursive path depth exceeds the limit threshold, an exception is automatically thrown and decentralized consensus chaining of the registration transaction is blocked;

2. Fixed-proportion asymmetric cascaded clearing and bookkeeping: upon receiving a data packet of a commercial settlement event, intercept an asset liquidation request containing a transaction amount **R**, split the amount **R** into a first proportional value retained by a current node, a second proportional value allocated to a parent protocol layer, and a third proportional value allocated to a grandparent protocol layer that satisfy a preset asymmetric descending gradient relationship, and write each proportional value into corresponding pending withdrawal balance state mapping variables of contracts in a ledger state tree in parallel in the form of non-immediate transfer increments;

3. Dynamic weight allocation: when the sub-protocol node inherits multiple parents, read self-consistency similarity between feature vector fingerprints of each parent protocol, calculate allocation weights proportionally, and accordingly dynamically adjust pending withdrawal balance increments of corresponding parent creator accounts in the state variables;

4. Expiry judgment and redirection bookkeeping: store a creation timestamp of each protocol node in the ledger state tree; when allocating pending split amounts to a target node, if a difference between a current consensus timestamp and the creation timestamp of the node exceeds a set protection period time threshold, or the node is determined as a root-level meta-protocol with zero private revenue, trigger a contract state machine transition to forcibly redirect and accumulate the split amount corresponding to the node into a pending withdrawal balance variable of an ecological directional backfeed pool contract account;

5. External pull-type physical withdrawal: respond to a transfer transaction initiated by a payee, verify pending withdrawal balances under the initiator’s name in the state variables, clear the pending withdrawal balance state variable, and then trigger an underlying on-chain transfer instruction to execute physical transfer to the payee, so as to physically isolate and eliminate reentrancy calls.

Claim 5

The method for automatic multi-level revenue allocation of cognitive assets according to **Claim 4**, characterized in that:
If a sub-protocol node inherits multiple parent protocols {F_1, F_2, ..., F_n}, the smart contract reads fingerprint similarity feature scores S_i between feature vector fingerprints of each registered parent protocol and the sub-protocol node in the ledger runtime state, and calculates allocation weights via the formula below:

```
Weight_i = (S_i) / (∑_j=1^n S_j)
```

The smart contract dynamically adjusts pending withdrawal balance increments of corresponding parent creator accounts in the state variables according to the allocation weights; if there is no similarity difference, the amount at this layer is split equally.

Claim 6

The method for automatic multi-level revenue allocation of cognitive assets according to **Claim 4**, characterized in that:
In the step of external pull-type physical withdrawal, the contract conducts fund transfer via a strict **two-phase atomic sequence**: at the entry of the withdrawal transaction, the pending withdrawal balance state variable of the caller is forcibly reset to zero within the consensus execution context to complete state tree update, and then an underlying on-chain transfer instruction is invoked to execute token transfer to the external wallet of the caller; if the transfer instruction returns an execution failure signal, a contract transaction rollback is triggered to restore the pending withdrawal balance state variable, so as to physically isolate concurrent reentrancy execution paths of external reentrancy calls by priority state modification.

Claim 7

A ledger contract-side operation method in the method for automatic multi-level revenue allocation of cognitive assets based on smart contracts, executed by a smart contract clearing module deployed on a decentralized ledger node, characterized by comprising the following steps:
1. Lineage depth registration access control: upon receiving a sub-protocol node registration

transaction, read parent protocol hash fingerprints declared in its data structure, recursively construct a derived topology graph along version dependency records; if the recursive path depth exceeds the limit threshold of **18 levels**, reject registration and trigger blocking of consensus chaining;

2. Transaction balance cascaded increment calculation: upon receiving an asset liquidation request containing a transaction amount **R**, split the amount **R** into a first proportional value retained by a current node, a second proportional value allocated to a parent protocol layer, and a third proportional value allocated to a grandparent protocol layer that satisfy an asymmetric descending gradient relationship, and write each proportional value into pending withdrawal balance state mapping variables of the ledger state tree in parallel in the form of increments;

3. Multi-parent similarity weighting and redirection: if multiple parent protocols are inherited, read their fingerprint similarity feature scores to calculate allocation weights and dynamically update corresponding pending withdrawal balance increments; if the difference between the creation timestamp of a target node and the current consensus timestamp reaches **20 years**, or the node is a root-level meta-protocol with zero private revenue, trigger state transition to redirect and accumulate the corresponding split amount into a pending withdrawal balance variable of the public backfeed pool contract account.

Claim 8

A creator-side revenue extraction method in the method for automatic multi-level revenue allocation of cognitive assets based on smart contracts, executed by a creator payment terminal device, characterized by comprising the following steps:

1. Account pending balance query: send a pending withdrawal balance query request from a local terminal to the multi-level allocation smart contract deployed on the decentralized ledger, to obtain values of pending withdrawal balance state mapping variables accumulated via cascaded allocation under the current account;

2. Pull-type withdrawal transaction signature initiation: use the creator’s asymmetric private key to digitally sign a withdrawal request containing the withdrawal amount and payment account, and initiate a withdrawal transaction to the smart contract, so as to trigger deduction and anti-reentrancy transfer verification based on a two-phase atomic sequence at the smart contract side;

3. Clearing and settlement fund receipt: after the smart contract side passes verification, clears and updates the pending withdrawal balance state mapping variable, and invokes an underlying

transfer instruction to complete physical transfer, receive the asset balance transferred on-chain at the local terminal.

Claim 9

The ledger contract-side operation method according to **Claim 7**, characterized in that:
The step of transaction balance cascaded increment calculation further includes a timing sequence verification sub-step: read timestamps in the liquidation request and verify timing sequence rationality to defend against replay attacks targeting the asset liquidation request; in multi-thread concurrent processing of ledger nodes, add concurrent control state identifiers based on timestamps and owner public keys to pending withdrawal balance state mapping variables corresponding to protocol nodes.

Claim 10

The creator-side revenue extraction method according to **Claim 8**, characterized in that:
1. In the step of account pending balance query, the pending withdrawal balance query request is sent to the smart contract via an asymmetric encrypted channel, and returned values of pending withdrawal balance state mapping variables are protected via public key encryption;
2. In the step of pull-type withdrawal transaction signature initiation, mix the latest block hash provided by the decentralized ledger or a one-time random **Nonce** during digital signature, to physically block third-party interception of withdrawal requests for withdrawal replay attacks.

Claim 11

A computer-readable storage medium storing a computer program thereon, characterized in that the program, when executed by a processor, implements the method according to any one of **Claim 4** to **Claim 10**.